

Class Test 1 Solution

Q1

```
1 Read x, Y
2 if (x > 0)
3   print "A"
4   if (Y > 0)
5     print "B"
6 else
7   print "C"
8 print "D"
```

Questions

- ① For the test case $x = -2, Y = 5$
- ② Calculate
 - I) Statement Coverage (%)
 - II) Decision Coverage (%)
- ③ Identify the Statements not executed
- ④ Suggest additional test cases to achieve 100% Decision Coverage

⇒

Total Statements = 4

- ① Print "A"
- ② Print "B"
- ③ Print "C"
- ④ Print "D"

* Test case Given

$x = -2, Y = 5$

* Execution Path for given Test case

1 → 2 (False) → 7 → 8

Executed Statement

- ① Print "C"
- ② Print "D"

I) Statement Coverage (%)

Total Statement = 4

Statement Executed for given test case = 2

Formula for Statement Coverage

$$\text{Statement Coverage} = \frac{\text{No. of executed Statement}}{\text{Total Number of Statement}} \times 100$$

$$= \frac{2}{4} \times 100$$

Statement coverage = 50%

II) Decision Coverage (-1.1)

Total Decision = 4

- ① if ($x > 0$) $\rightarrow$ True
- ② if ($x > 0$) $\rightarrow$ False
- ③ if ($y > 0$) $\rightarrow$ True
- ④ if ($y > 0$) $\rightarrow$ False

Test case given $x = -2$ $y = 5$

Only one Decision cover if ($x > 0$) $\rightarrow$ False

$\therefore$ cover only one decision

Formula for Decision coverage

$$\text{Decision coverage} = \frac{\text{No. of Decision executed}}{\text{Total No. of Decision}} \times 100$$

$$= \frac{1}{4} \times 100$$

$\text{Decision coverage} = 25\%$

③ Statement Not Executed

- ① Print "A"
- ② Print "B"

④ Additional Test cases for 100% Decision Coverage

① Tc 1 = $x = 5$, $y = 5$
cover $x > 0 \rightarrow$ True
 $y > 0 \rightarrow$ False

② Tc 2 = $x = 5$, $y = -5$
cover $x > 0 \rightarrow$ True
 $y > 0 \rightarrow$ False

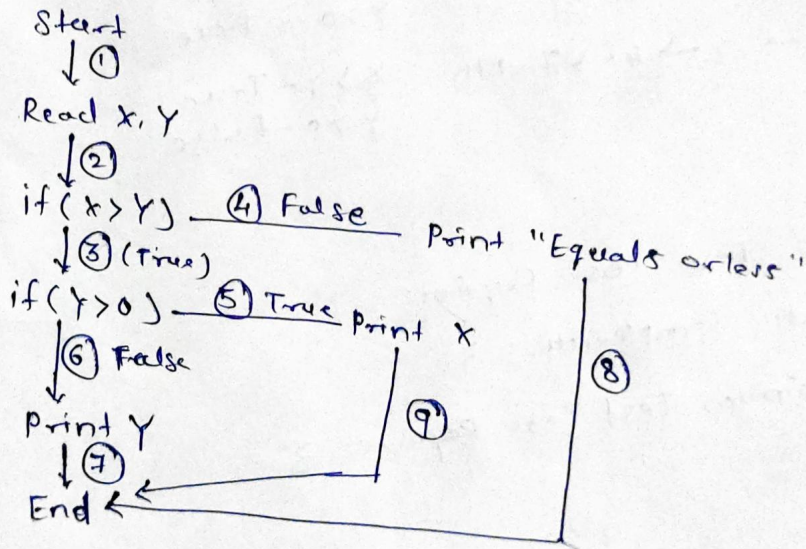
Q2

- 1 Start
- 2 Read x, y
- 3 if (x > y)
- 4 if (y > 0)
- 5 Print x
- 6 else
- 7 Print y
- 8 else
- 9 Print "Equal or less"
- 10 End.

Question

- ① Draw the Control Flow Graph (CFG) for the Program
- ② Calculate Cyclomatic Complexity using
 - (a) Edge-Node Method
 - (b) Decision Point method
- ③ List All Independent Path
- ④ Determine the minimum number of test cases required

⇒ ① CFG



Total Edges = 9
Nodes = 8

② (a) Cyclomatic complexity using Edge-Node Method

Formula

$$\begin{aligned}
 V(G) &= E - N + 2P \\
 &= 9 - 8 + 2 \times 1 \\
 &= 1 + 2
 \end{aligned}$$

cyclomatic complexity = 3

(b) Using Decision Point Method

Decision Node

$X > Y$

$Y > 0$

Formula

$$\begin{aligned} V(G) &= D + 1 \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

③ Independent Paths.

1. $1 \rightarrow 2 \rightarrow 3 \rightarrow 9 \rightarrow 10$ $(X > Y \rightarrow \text{False}) \rightarrow \text{Path}$

2. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 10$ $X > Y - \text{True}$
 $Y > 0 - \text{True}$

3. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 7 \rightarrow 10$ $X > Y - \text{True}$
 $Y > 0 - \text{False}$

④ Minimum Test case Require

Cyclomatic complexity = 3

So Minimum Test case Require = 3

- Q3 An online banking System accept a withdrawal amount with following conditions:
- Amount Per day withdraw Rs. 100 to Rs. 10,000
 - Amount Must be in multiplier of Rs. 100

Questions

1. Identify all Equivalence classes (Valid & Invalid)
2. Design Test cases using Equivalence Partitioning
3. Identify the boundary value for the input
4. Design test case using Boundary value Analysis

→

① Equivalence classes

Valid classes

- (i) 100 to 10,000
- (ii) value Multiplier of 100

Invalid classes

- (i) less than 100
- (ii) Greater than 10,000
- (iii) Not multiplier of 100

② Test cases

Test case Id	Input Amount	Partition	Expected Result
Tc 1	500	valid (100-10,000)	Amount Accepted
Tc 2	205	Invalid (NOT Multiplier of 100)	Amount Rejected
Tc 3	95	Invalid (<100)	Reject Amount
Tc 4	15000	Invalid (>10,000)	Amount Greater than limit

③ Boundary value for Input

Minimum value = 100

Maximum value = 10,000

Boundary values

99, 100, 101

9999, 10,000, 10001

④ Test cases

Test case Id	Input Amount	Boundary condition	Expected Result
Tc1	99	Min - 1	Reject Amount
Tc2	100	Min	Accept
Tc3	101	Min + 1	Reject (Not Multiple of 100)
Tc4	9999	Max - 1	Reject h
Tc5	10000	Max	Accept
Tc6	10001	Max + 1	Reject (Greater than Limit)